

Exact maximal order types below ω^2

A proposed solution to Altman's finite-alphabet sharpness question

Research draft prepared with ChatGPT

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Abstract

For a finite poset P , consider all words of ordinal length less than ω^2 , ordered by increasing subsequence embeddings that may increase labels in P . Write $n = |P|$ and let $j(P)$ count all downsets of P , including the empty one. This article gives a complete proposed proof that, for $n \geq 2$,

$$o(s_{\omega^2}^F(P)) = \omega^{\omega^{n+j(P)-2}}.$$

In particular, the two-letter antichain has maximal order type ω^{ω^4} , answering the explicit sharpness question in Example 3.15 of Altman's March 2026 revision. The two-letter chain has type ω^{ω^3} . The main construction embeds every finite Cartesian power of an earlier stage into the next stage by synchronizing periodic ω -blocks. Proper downsets admit an excluded-letter guard; the universal block instead requires a limit-length padding argument. A more general classification for selected families of periodic tails isolates the exact exception in which naive block counting fails. An exact symbolic embedding algorithm and a reproducible test suite accompany the proofs.

Status and scope. This is a new, unrefereed research argument, not a report of an independently certified result. All proof steps beyond the explicitly stated classical maximal-order-type theorems are supplied below. The computer checks test symbolic embeddings, not infinite ordinal ranks. The claim concerns finite alphabet posets and the strict cutoff ω^2 ; it does not settle arbitrary alphabets or higher cutoffs.

Keywords. Well-quasi-order; maximal order type; transfinite word; Higman ordering; finite downset lattice; natural product.

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1 The selected question and the result

Altman studies finite-image words of bounded transfinite length in [1]. Example 3.15, on printed page 10 of version 2, gives the upper bound ω^{ω^4} for a two-element antichain and explicitly leaves its sharpness unverified. The same example improves the chain's upper bound to ω^{ω^3} . The arXiv record checked for this note lists version 2, dated March 10, 2026, as the latest revision. Targeted searches did not locate a subsequent resolution; this is a limited literature check, not a guarantee of priority.

This is an attractive problem to attack because the alphabet is tiny, all ω -tails have finite descriptions up to embedding, and the candidate ordinal is explicit. The difficult part is not finding more tail types. It is proving that the types already available contribute their full ordinal complexity after concatenation.

Let P be a finite poset. A *downset* is a subset $D \subseteq P$ such that $x \leq_P y \in D$ implies $x \in D$. Put

$$\mathcal{J}(P) = \{D \subseteq P : D \text{ is a downset}\}, \quad j(P) = |\mathcal{J}(P)|.$$

The empty set is counted in $j(P)$ throughout.

Theorem 1.1 (Finite-alphabet formula). *Let P be a finite poset with $n \geq 2$ elements. With increasing, label-respecting subsequence embedding and mutual embeddability quotiented out,*

$$o(s_{\omega^2}^F(P)) = \omega^{\omega^{n+j(P)-2}}. \quad (1.1)$$

For the empty alphabet the maximal order type is 1, and for a singleton alphabet it is ω^2 .

Here $s_{\omega^2}^F(P)$ denotes words of length strictly less than ω^2 . Since P is finite, the finite-image superscript is automatic. The theorem is proved in Section 7 after two product embedding lemmas.

Corollary 1.2 (The selected binary cases). *For a two-element antichain A_2 and a two-element chain C_2 ,*

$$o(s_{\omega^2}^F(A_2)) = \omega^{\omega^4}, \quad o(s_{\omega^2}^F(C_2)) = \omega^{\omega^3}.$$

Proof. There are four downsets in A_2 and three in C_2 . Substitute $n = 2$ in (1.1). $\square$

The proof establishes more than these two values. We may choose an arbitrary family of nonempty downsets and allow only the corresponding periodic ω -blocks. Unless the universal block is the sole allowed infinite block, the number of allowed block types determines the maximal order type exactly. The exceptional case is essential, not a technical inconvenience.

1.1 What the proof adds

The upper bound follows from a finite generating alphabet. The lower bound requires an additional construction. A new periodic block serves as a separator only when an embedding cannot hide it inside earlier blocks. Introducing downsets in inclusion order solves that problem. A finite guard then prevents a component from leaking into the next separator. At the universal separator no excluded letter exists, so a different device is needed: attach a cancellable tail of limit length. An infinite tail cannot fit into a bounded, hence finite, initial segment of an ω -block.

Together these observations give actual order embeddings

$$X^{r+1} \hookrightarrow Y \quad (r < \omega)$$

from each stage into the next. Classical natural-product arithmetic turns these embeddings into the sharp lower bound.

2 Definitions and classical inputs

2.1 Ordinal words and embeddability

A *word over P* is a function $u : \alpha \rightarrow P$, where α is an ordinal; its length is $\text{len}(u) = \alpha$. If $u : \alpha \rightarrow P$ and $v : \beta \rightarrow P$, write $u \preccurlyeq v$ when there is a strictly increasing map $f : \alpha \rightarrow \beta$ satisfying

$$u(\xi) \leq_P v(f(\xi)) \quad (\xi < \alpha).$$

No continuity or cofinality condition is imposed on f . Write $u \simeq v$ when $u \preccurlyeq v$ and $v \preccurlyeq u$. Concatenation is ordinal concatenation. The empty word ε has length zero. For lengths below ω^2 there are unique $k, m < \omega$ such that

$$\text{len}(u) = \omega k + m.$$

Embeddability is in general only a quasi-order. Distinct words, such as a^ω and aa^ω , may be equivalent. All maximal order types in this article refer to the quotient by $\simeq$.

Lemma 2.1 (Concatenation is monotone). *If $u_i \preccurlyeq v_i$ for $0 \leq i < r$, where r is finite, then $u_0 \cdots u_{r-1} \preccurlyeq v_0 \cdots v_{r-1}$.*

Proof. Translate the embedding on each component into the corresponding interval of the target concatenation. The intervals occur in the same order, so the union of these translated maps is strictly increasing and respects labels. $\square$

A finite concatenation of words of length below ω^2 still has length below ω^2 . In particular, all maps used later stay within the stated cutoff, even though the number of ω -blocks in their arguments is unbounded across different arguments.

2.2 Maximal order types

A quasi-order is a *well-quasi-order* if every infinite sequence $x_0, x_1, \dots$ contains $i < j$ with $x_i \leq x_j$. The maximal order type $o(X)$ is the largest ordinal type of a linear extension of the quotient poset. The existence of this maximum, its characterization by the tree of bad finite sequences, and the product theorem below are classical [2]. We use the following precise inputs.

Theorem 2.2 (Classical maximal-order-type calculus). *For well-quasi-orders X, Y :*

- (i) *An order embedding $X \hookrightarrow Y$ implies $o(X) \leq o(Y)$.*
- (ii) *An order-preserving surjection from Y onto X , including surjectivity up to quasi-order equivalence, implies $o(X) \leq o(Y)$.*
- (iii) *For the componentwise Cartesian product, $o(X \times Y) = o(X) \otimes o(Y)$, where $\otimes$ is natural product.*
- (iv) *If P is a nonempty finite poset of cardinality n , its Higman word ordering satisfies*

$$o(P^*) = H_n, \quad H_n := \omega^{\omega^{n-1}}. \quad (2.1)$$

The finite-word formula is part of the de Jongh–Parikh–Schmidt theory [2, 3]. These are imported theorems; the present article does not claim new proofs of them. All later lower bounds are reduced to these inputs by explicit embeddings.

For clarity, assertion (ii) has an elementary bad-sequence explanation. Choose a preimage for each element of the target quotient. If a target sequence is bad, its sequence of chosen preimages is bad, because any comparison in the source would project to a comparison in the target. This embeds the target bad-sequence tree into the source tree. The same argument shows the target is a well-quasi-order whenever the source is.

2.3 The ordinal arithmetic used here

For ordinals in Cantor normal form, natural sum $\oplus$ adds coefficients at equal exponents, and natural product $\otimes$ multiplies the resulting formal polynomials, using natural sum on exponents. In particular,

$$(\omega^\alpha) \otimes (\omega^\beta) = \omega^{\alpha \oplus \beta}.$$

Consequently, for finite $N \geq 1$ and $q \geq 1$,

$$H_N^{\otimes q} = \omega^{\omega^{N-1}q}, \quad (2.2)$$

where the multiplication by q on the right is ordinary ordinal multiplication by a positive integer. Continuity of ordinal exponentiation gives

$$\sup_{q \geq 1} H_N^{\otimes q} = \omega^{\sup_{q \geq 1} \omega^{N-1}q} = \omega^{\omega^N} = H_{N+1}. \quad (2.3)$$

Equation (2.3) is the entire ordinal calculation behind the induction. Ordinary powers and natural powers must not be interchanged in general; (2.2) states exactly the special case being used.

3 Periodic tails and the finite generating alphabet

3.1 One periodic block for each nonempty downset

For a nonempty downset $D = \{d_0, \dots, d_{t-1}\}$, fix a periodic word

$$B_D = (d_0 \cdots d_{t-1})^\omega. \quad (3.1)$$

The chosen enumeration is immaterial up to $\approx$: every element of D appears infinitely often in every such representative.

Lemma 3.1 (Comparison of periodic blocks). *For nonempty downsets D, E and a letter $a \in P$,*

$$a \preceq B_D \iff a \in D, \quad B_D \preceq B_E \iff D \subseteq E.$$

Every nonempty tail of B_D is equivalent to B_D .

Proof. If $a \leq_P d$ for a letter $d \in D$, downward closure gives $a \in D$. Conversely, $a \in D$ supplies a matching occurrence in B_D . If $B_D \preceq B_E$, applying the same observation to an occurrence of each $d \in D$ gives $D \subseteq E$. If $D \subseteq E$, recursively match the letters of B_D to successive identical occurrences in B_E . There is always a later occurrence, so the recursion gives an embedding. The identical recursive construction works in any nonempty tail of a periodic block; the reverse embedding into the full block is inclusion. $\square$

For a nonempty subset $R \subseteq P$, repeating R or repeating its downset $\downarrow R$ gives equivalent ω -words. In the less immediate direction, each $x \in \downarrow R$ can be matched to an element $r \in R$ above it, and each such r recurs infinitely often.

3.2 Reduction of arbitrary words

The following periodic-tail reduction is standard; compare [1, Propositions 3.3–3.4]. A proof is included to specify exactly which equivalence is needed.

Lemma 3.2 (Finite prefix and recurrent tail). *Every word $u : \omega \rightarrow P$ is equivalent to vB_D for some finite word $v \in P^*$ and some nonempty downset D .*

Proof. Let R be the set of labels occurring infinitely often in u . Since P is finite and u is infinite, R is nonempty. Every label outside R occurs only finitely often. Choose an index after all such occurrences, and write $u = vt$, where v is finite and t uses only labels from R . Every element of R occurs infinitely often in t .

Put $D = \downarrow R$. To embed t into B_D , successively choose later identical occurrences of its labels. To embed B_D into t , for each requested label $d \in D$ choose an $r \in R$ with $d \leq_P r$, and then choose an occurrence of r later than all previous choices. Infinitely many occurrences of r make this possible. Thus $t \approx B_D$, and monotonicity of concatenation gives $u \approx vB_D$. $\square$

Proposition 3.3 (Finite representation). *Every word of length less than ω^2 is equivalent to a finite concatenation of one-letter words and blocks B_D , with $\emptyset \neq D \in \mathcal{J}(P)$.*

Proof. Split a word of length $\omega k + m$ into its k consecutive intervals of length ω and its final finite interval. Apply Lemma 3.2 to each infinite interval and concatenate the resulting representatives. $\square$

This is a representation theorem, not a uniqueness theorem. Different finite expressions can represent equivalent words. For example, $aB_{\downarrow a} \approx B_{\downarrow a}$. Such absorption is precisely why a finite generating alphabet gives only an upper bound without further work.

Definition 3.4 (Selected families). For $\mathcal{F} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$, let $\mathcal{W}(P, \mathcal{F})$ be the quasi-order of all finite concatenations of letters in P and periodic blocks B_D with $D \in \mathcal{F}$, using actual ordinal-word embeddability. Finite expressions are identified up to mutual embeddability when maximal order type is taken.

In particular, $\mathcal{W}(P, \emptyset) = P^*$, and $\mathcal{W}(P, \mathcal{J}(P) \setminus \{\emptyset\})$ has the same quotient order as $s_{\omega^2}^F(P)$ by Proposition 3.3.

Proposition 3.5 (Atom-count upper bound). For $n = |P| \geq 1$ and $m = |\mathcal{F}|$,

$$o(\mathcal{W}(P, \mathcal{F})) \leq H_{n+m}. \quad (3.2)$$

Moreover, $\mathcal{W}(P, \mathcal{F})$ is a well-quasi-order.

Proof. Take a discrete alphabet of $n + m$ symbols: one for each finite letter and one for each selected periodic block. Evaluate a finite symbol string by ordinal concatenation. If one string is a subsequence of another, the corresponding evaluated word embeds into the other by Lemma 2.1. Evaluation is surjective onto the represented quasi-order. Apply the finite-word formula and monotone-surjection principle in Theorem 2.2. $\square$

Using a discrete alphabet deliberately ignores extra comparisons among the atoms. This does not weaken the desired upper bound: every poset with the same positive finite cardinality has the same finite-word maximal order type. No order-reflection claim is made for the evaluation map.

4 Cofinal localization and synchronization

The following lemma concerns *displayed target tokens*. A finite word may contain many one-letter tokens adjacent to each other, and a token expression need not be a canonical decomposition of its ordinal length. Nevertheless, each displayed B_E occupies a convex interval of order type ω , and there are only finitely many displayed tokens.

Lemma 4.1 (Cofinal localization). Suppose a periodic block B_D embeds into a finite token expression v . Then a tail of its image lies in one displayed target block B_E , is cofinal in that block, and satisfies $D \subseteq E$. If a source position occurs after the whole source block B_D , its image occurs after the whole target block B_E .

Proof. Number the target tokens in their order of occurrence. As the source positions $0, 1, 2, \dots$ increase, the indices of the tokens containing their images form a nondecreasing sequence with only finitely many possible values. It is eventually constant. The final token cannot be a one-letter token, because the embedding is injective. It is therefore some B_E . The image contains infinitely many positions of that token, so it is unbounded in its order type ω .

Every $d \in D$ still occurs in the source tail. Each such occurrence maps to a label in E above d . Downward closure gives $d \in E$, proving $D \subseteq E$. Finally, the image of any subsequent source position must lie above every image point of B_D . Since those points are cofinal in the target B_E , no position of that target block remains available. $\square$

Remark 4.2 (A block may straddle earlier blocks). The lemma does not say that the entire source block lands in a single target block. An initial finite part can be placed earlier. Only the tail is localized. All subsequent proofs use this weaker, correct statement.

Lemma 4.3 (Equal-count synchronization). Let D be a nonempty downset, and suppose an old family $\mathcal{G}$ satisfies

$$D \not\subseteq E \quad \text{for every } E \in \mathcal{G}. \quad (4.1)$$

Consider two token expressions with exactly r displayed copies of B_D , all other infinite tokens belonging to $\mathcal{G}$. Under an embedding of the first expression into the second, the tail of the i th displayed source B_D is cofinal in the i th displayed target B_D , for $1 \leq i \leq r$.

Proof. By Lemma 4.1, each displayed source B_D requires a target infinite block whose downset contains D . Condition (4.1) excludes all old target blocks. Its cofinal destination is therefore one of the r displayed new target blocks. Successive source new blocks have strictly increasing cofinal destination indices: after the first has used a target block cofinally, any later source position must be beyond it. A strictly increasing injection from $\{1, \dots, r\}$ into itself is the identity. This proves the claim. $\square$

To arrange (4.1), introduce a finite collection of downsets in an inclusion-respecting order. Ordering them by cardinality, with arbitrary tie breaking, suffices: a new set cannot be contained in an earlier, distinct set of no larger size. The full downset P , when present, is last.

5 Proper markers: finite guards give product embeddings

Lemma 5.1 (Right cancellation of a final letter). *For arbitrary ordinal words u, v and any fixed letter c ,*

$$uc \preceq vc \implies u \preceq v.$$

Proof. The image of the final source letter lies at or before the final target letter. Every position belonging to u maps strictly before that image, hence inside the target prefix v . Restrict the embedding. $\square$

Theorem 5.2 (Proper-marker product embedding). *Let D be a nonempty proper downset of P , and let $\mathcal{G}$ satisfy (4.1). Choose $c \in P \setminus D$ and put $X = \mathcal{W}(P, \mathcal{G})$, $Y = \mathcal{W}(P, \mathcal{G} \cup \{D\})$. For every $r \geq 1$, the map*

$$\Phi_r(u_0, \dots, u_r) = u_0 c B_D u_1 c B_D \cdots u_{r-1} c B_D u_r \quad (5.1)$$

is an order embedding $X^{r+1} \hookrightarrow Y$.

Proof. Order preservation follows by concatenating componentwise embeddings and identity maps on the separators. It remains to prove reflection.

Suppose $\Phi_r(u_0, \dots, u_r) \preceq \Phi_r(v_0, \dots, v_r)$. By Lemma 4.3, the i th source separator B_D has image tail cofinal in the i th target separator B_D . Thus, for $i > 0$, every image point of u_i and of its subsequent guard lies after the i th target separator. For $i = 0$ there is no preceding separator and no lower-bound restriction to impose.

Fix $i < r$. The source guard c immediately following u_i precedes the next source separator. Some point of that next separator maps into the corresponding next target separator. Therefore the source guard's image is before the end of that target separator. It cannot be in that separator: if it were matched to a label $d \in D$, then $c \leq_P d$ would imply $c \in D$, contrary to the choice of c . Its image consequently lies in the target interval $v_i c$ before the separator. Every point of u_i precedes the guard, so its image lies in v_i , not in the final guard of $v_i c$. We obtain $u_i \preceq v_i$.

After the last source separator, cofinality forces the entire source suffix u_r to map into the target suffix v_r . Hence $u_r \preceq v_r$ as well. All components compare in the product order. Reflection and preservation imply an order embedding after taking equivalence classes. $\square$

The guard is allowed to map into the interior of v_i ; it need not match the explicitly displayed target guard. The proof only needs the fact that it lies no later than that guard. This is important when old components already contain many occurrences of c or old infinite blocks.

Corollary 5.3 (One proper marker raises the level). *Under the hypotheses of Theorem 5.2, if $o(X) = H_N$ and $n + |\mathcal{G}| = N$, then $o(Y) = H_{N+1}$.*

Proof. For every $r \geq 1$, product embedding and the classical product theorem give $o(Y) \geq H_N^{\otimes(r+1)}$. Taking the supremum and using (2.3) yields $o(Y) \geq H_{N+1}$. Proposition 3.5 gives the reverse inequality. $\square$

6 The universal marker: limit padding replaces a guard

For $D = P$ there is no excluded letter. Any fixed finite word z is absorbed by the universal periodic block:

$$zB_P \simeq B_P. \quad (6.1)$$

Indeed, the countable sequence zB_P can be matched greedily into B_P ; every requested label reappears infinitely often. Thus attaching a finite guard before B_P cannot preserve arbitrary component information. The remedy is to end the component with a cancellable *infinite* tail.

6.1 A cancellable map into limit lengths

Lemma 6.1 (Limit padding). *Let E be a nonempty proper downset of P , and choose $c \notin E$. The map*

$$G(u) = ucB_E \quad (6.2)$$

is order preserving and order reflecting on arbitrary words of length below ω^2 . Every $G(u)$ has nonzero limit length, of the form ωk with $k \geq 1$. If B_E is allowed in a family, G maps that family's represented word order into itself.

Proof. Monotonicity is immediate. Suppose $ucB_E \preccurlyeq vcB_E$. The source occurrence of c cannot map into the final target B_E , by downward closure and $c \notin E$. Thus it maps into vc , and the restriction to u maps into v , by the same final-letter argument as Lemma 5.1. Hence $u \preccurlyeq v$.

If $\text{len}(u) = \omega k + m$, then

$$\text{len}(ucB_E) = (\omega k + m) + 1 + \omega = \omega(k + 1),$$

which is a nonzero limit ordinal. The family assertion follows from the explicit concatenation. $\square$

Lemma 6.2 (No infinite tail fits below a finite bound). *Let u have nonzero limit length below ω^2 . Suppose an embedding maps all of u to positions preceding a fixed position η inside a target interval of order type ω . Then no position of u can map inside that interval.*

Proof. If a source position ξ mapped into the interval before η , there would be only finitely many target positions between its image and η . But a nonzero limit ordinal has infinitely many positions after each of its positions: otherwise the ordinal would have a last position. The infinitely many source positions after ξ would have to inject into that finite target interval. This is impossible. $\square$

The position η need not be the first position of the target ω -block. Any point of it gives a finite bound. Nor is it necessary for the next source marker to begin inside that block: one image point of its cofinal tail suffices.

6.2 The product embedding

Theorem 6.3 (Universal-marker product embedding). *Let $\mathcal{G}$ consist of nonempty proper downsets of P , with $\mathcal{G} \neq \emptyset$. Choose $E \in \mathcal{G}$ and the map G from Lemma 6.1. Put $X = \mathcal{W}(P, \mathcal{G})$ and $Y = \mathcal{W}(P, \mathcal{G} \cup \{P\})$. For each $r \geq 1$,*

$$\Psi_r(u_0, \dots, u_r) = G(u_0)B_P G(u_1)B_P \cdots G(u_{r-1})B_P u_r \quad (6.3)$$

is an order embedding $X^{r+1} \hookrightarrow Y$.

Proof. Preservation follows from the monotonicity of G and concatenation. Suppose the image of $(u_0, \dots, u_r)$ embeds into the image of $(v_0, \dots, v_r)$.

No old downset contains P . Lemma 4.3 therefore applies to the r displayed universal separators. Their cofinal destinations are the corresponding r displayed target universal separators.

Fix $i < r$. When $i > 0$, cofinality at the preceding separator forces all of $G(u_i)$ to map after the corresponding preceding target separator. Choose an image point η of the following source B_P inside the corresponding following target B_P ; such a point exists by cofinal localization. Every image point of $G(u_i)$ is before η . By Lemma 6.1, $G(u_i)$ has nonzero limit length. Applying Lemma 6.2 shows that none of its positions can be in this following target B_P . Thus the whole image of $G(u_i)$ lies in the intervening target component $G(v_i)$. For $i = 0$ the same argument applies without a preceding boundary. We have proved

$$G(u_i) \preccurlyeq G(v_i) \quad (i < r).$$

Reflection for G yields $u_i \preccurlyeq v_i$ for all $i < r$.

After the last source universal separator, cofinality forces the final source component u_r into the final target component v_r . Hence $u_r \preccurlyeq v_r$, completing product reflection. $\square$

Corollary 6.4 (One universal marker raises the level when padding exists). *Under the hypotheses of Theorem 6.3, if $o(X) = H_N$ and $n + |\mathcal{G}| = N$, then $o(Y) = H_{N+1}$.*

Proof. Apply the finite Cartesian-product theorem to the embeddings Ψ_r , take the supremum in (2.3), and combine with the atom-count upper bound, exactly as in Corollary 5.3. $\square$

Notice the role of the old proper block. It need not encode the whole component by itself. Its job is to give a limit-length ending, while an excluded finite letter makes that ending cancellable. These two properties are logically distinct and both are used.

7 The full classification and the main theorem

Theorem 7.1 (Selected-tail classification). *Let P be a nonempty finite poset of size n , and let $\mathcal{F} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$.*

(i) *If $\mathcal{F} \neq \{P\}$, then*

$$o(\mathcal{W}(P, \mathcal{F})) = H_{n+|\mathcal{F}|} = \omega^{\omega^{n+|\mathcal{F}|-1}}. \quad (7.1)$$

This includes the empty family.

(ii) *If $\mathcal{F} = \{P\}$, then*

$$o(\mathcal{W}(P, \{P\})) = H_n \cdot \omega = \omega^{\omega^{n-1}+1}. \quad (7.2)$$

Proof of part (i). For the empty family, (7.1) is the finite-word formula. Otherwise $\mathcal{F} \neq \{P\}$ implies that the family contains at least one proper downset. List its proper downsets as $D_1, \dots, D_t$ in increasing cardinality, breaking ties arbitrarily. If $P \in \mathcal{F}$, append it last.

Start with $X_0 = P^*$, of type H_n . At step $i \leq t$, no earlier downset contains D_i . Indeed, an earlier downset has no greater cardinality, and equality together with containment would make the downsets equal. Theorem 5.2 applies. Inductively, Corollary 5.3 gives

$$o(\mathcal{W}(P, \{D_1, \dots, D_i\})) = H_{n+i}.$$

If P is absent from $\mathcal{F}$, this finishes the proof. If P is present, there is an old proper downset because $t \geq 1$. Apply Theorem 6.3 and Corollary 6.4 to the final step, obtaining $H_{n+t+1} = H_{n+|\mathcal{F}|}$. $\square$

Proof of part (ii). Every expression using only the universal infinite block can be written

$$u_0 B_P u_1 B_P \cdots u_{k-1} B_P v, \quad u_i, v \in P^*.$$

By (6.1), it is equivalent to $B_P^k v$. Here B_P^k means k successive copies, not an infinite repetition. These normal forms satisfy

$$B_P^k v \preccurlyeq B_P^\ell w \iff (k < \ell) \text{ or } (k = \ell \text{ and } v \preccurlyeq w). \quad (7.3)$$

If $k < \ell$, send the k source infinite blocks into the first k target blocks and the finite suffix into one extra universal target block. If $k > \ell$, k successive source infinite blocks require k distinct cofinal target infinite destinations by Lemma 4.1, which is impossible. If $k = \ell$, synchronization leaves the source suffix wholly in the target suffix, so $v \preccurlyeq w$ is necessary. It is plainly sufficient.

Thus the quotient is the ordered sum of ω copies of P^* :

$$\mathcal{W}(P, \{P\})/\simeq \cong \sum_{k < \omega} P^*. \quad (7.4)$$

Every linear extension must put all of the k th copy before the $(k+1)$ st copy. Each copy has maximal order type H_n , and one may use a maximal linear extension in every copy. Therefore the maximal order type is the ordinary ordered sum $\sum_{k < \omega} H_n = H_n \cdot \omega$. The elementary identity $\omega^\alpha \cdot \omega = \omega^{\alpha+1}$ gives (7.2). $\square$

Proof of Theorem 1.1. Suppose $n \geq 2$. Choose a minimal element a of P . The singleton $\{a\}$ is a nonempty proper downset. Consequently the full family $\mathcal{J}(P) \setminus \{\emptyset\}$ is not the exceptional family $\{P\}$. Its size is $j(P) - 1$. Proposition 3.3 and Theorem 7.1(i) give

$$o(s_{\omega^2}^F(P)) = H_{n+j(P)-1} = \omega^{\omega^{n+j(P)-2}}.$$

For $n = 1$, there is one word at each ordinal length $\alpha < \omega^2$, and embedding is exactly comparison of lengths. Its quotient is therefore the ordinal ω^2 itself. Equivalently, apply part (ii) with $H_1 = \omega$. For $n = 0$, the empty word is the sole word, giving maximal order type 1. $\square$

7.1 Why the supremum argument is legitimate

The proof never identifies the order type of an increasing union with the supremum of the order types of its pieces. Such an identification is not a general principle for maximal order types. Instead, for each fixed finite r , there is a separately verified order embedding of a Cartesian power. It gives a genuine lower bound on the *same* target order type. Taking the supremum of a collection of lower bounds is valid. The upper bound is independently obtained by a monotone surjection.

7.2 The singleton warning is structural

For a singleton alphabet, two formal atoms are available: the letter and its periodic ω -block. Their finite words would have maximal type $H_2 = \omega^\omega$ if treated as a free Higman alphabet. But after ordinal evaluation, the actual type is only ω^2 . The family $\mathcal{F} = \{P\}$ exhibits the same defect over every nonempty finite alphabet. Hence atom counting alone cannot be a lower-bound proof. The padding construction explains exactly why the full family over $n \geq 2$ avoids this collapse.

8 The binary proof written out explicitly

This section specializes the general proof so the selected problem can be checked without tracking an arbitrary downset family.

8.1 Two incomparable letters

Let $P = \{a, b\}$ with no comparisons between distinct letters, and abbreviate

$$A = a^\omega, \quad B = b^\omega, \quad U = (ab)^\omega.$$

There are exactly three nonempty downsets, giving these three periodic block types. Define stages by their allowed tokens:

stage	allowed tokens	claimed maximal type
X_0	a, b	$H_2 = \omega^\omega$
X_1	a, b, A	$H_3 = \omega^{\omega^2}$
X_2	a, b, A, B	$H_4 = \omega^{\omega^3}$
X_3	a, b, A, B, U	$H_5 = \omega^{\omega^4}$

The upper bound in each row comes from the number of tokens. The first equality is classical. To pass from X_0 to X_1 , use, for every finite r ,

$$(u_0, \dots, u_r) \mapsto u_0 b A u_1 b A \cdots u_{r-1} b A u_r.$$

Source A -blocks must synchronize with target A -blocks, and the guard b cannot enter A . This embeds X_0^{r+1} .

To pass from X_1 to X_2 , use

$$(u_0, \dots, u_r) \mapsto u_0 a B u_1 a B \cdots u_{r-1} a B u_r.$$

A source B -block cannot be cofinal in an old target A -block, and its preceding guard a cannot enter a target B -block. This embeds X_1^{r+1} .

For the final step, set

$$G(u) = u b A.$$

The guard b cannot enter the final A , so G reflects embedding; the image has limit length. Now use

$$(u_0, \dots, u_r) \mapsto G(u_0) U G(u_1) U \cdots G(u_{r-1}) U u_r.$$

A source U contains both letters infinitely often, so neither A nor B can be its cofinal destination. The displayed U -blocks synchronize. Limit padding prevents the intervening components from entering the next U before a fixed image point of its source marker. This embeds X_2^{r+1} .

At each of the three steps, (2.3) converts these product embeddings into the next displayed ordinal. Finally, every binary word of length below ω^2 is equivalent to an expression in a, b, A, B, U . This proves the sharpness of the binary antichain bound.

8.2 Two ordered letters

Let $0 < 1$. The nonempty downsets are $\{0\}$ and $\{0, 1\}$; their periodic blocks may be represented by

$$A = 0^\omega, \quad U = 1^\omega.$$

The latter is equivalent to $(01)^\omega$. Begin with finite words over $\{0, 1\}$, of maximal type H_2 . Add A using the guard 1, giving H_3 . Then add U using the limit padding $G(u) = u1A$, giving H_4 . Thus the chain's sharpened upper bound is also attained.

8.3 Three easily confused comparisons

For the antichain, the following facts illustrate the mechanism:

$$aU \simeq U, \quad A \preccurlyeq U, \quad AB \not\preccurlyeq U.$$

The first is finite-prefix absorption. The second uses cofinal recurrence of a in U . The third follows because two successive ω -blocks cannot both have cofinal image in the single target block. Also $U \not\preccurlyeq AB$: its tail would have to lie in either the pure a block or the pure b block, losing one of its recurrent labels. These facts show why both order of occurrence and cofinal type matter.

9 Consequences and finite-poset examples

9.1 Chains, antichains, and extremal values

Corollary 9.1. *For $n \geq 2$,*

$$o(s_{\omega^2}^F(C_n)) = \omega^{\omega^{2n-1}}, \quad o(s_{\omega^2}^F(A_n)) = \omega^{\omega^{n+2n-2}}.$$

Among n -element posets, the first value is the minimum, attained exactly by chains, and the second is the maximum, attained exactly by antichains.

Proof. A chain has exactly $n + 1$ downsets, its initial segments. Every subset of an antichain is a downset, so it has 2^n downsets. For any finite poset, the initial segments of a linear extension give $n + 1$ distinct downsets, and there are at most 2^n subsets in total.

For the equality statement at the lower end, fix a linear extension. If there are only $n + 1$ downsets, every downset is one of its initial segments. In particular all principal downsets are comparable by inclusion. Since $\downarrow x \subseteq \downarrow y$ implies $x \leq_P y$, every pair of elements is comparable, and P is a chain. At the upper end, if every subset is a downset, a strict relation $x <_P y$ would make $\{y\}$ fail to be a downset. Thus P is an antichain. The function of $j(P)$ in (1.1) is strictly increasing, so the ordinal extremal assertions follow. $\square$

9.2 Equivalent counting invariants

Every downset of a finite poset is generated by its maximal elements, which form an antichain. Conversely, the maximal elements of the downset generated by an antichain are that antichain. These maps give a bijection, including the empty set. Thus $j(P)$ may equally be described as the number of antichains of P .

Complementation gives a bijection between downsets and upsets. Passing to the dual poset exchanges downsets with upsets, so $j(P^{\text{op}}) = j(P)$. The formula therefore yields

$$o(s_{\omega^2}^F(P^{\text{op}})) = o(s_{\omega^2}^F(P)).$$

This is equality of maximal order types, not a claim that the word quasi-orders are isomorphic.

For a disjoint union $P \amalg Q$, choosing a downset is equivalent to choosing one downset in each component, hence $j(P \amalg Q) = j(P)j(Q)$. For an ordered sum $P + Q$, in which every element of P lies below every element of Q , one has $j(P + Q) = j(P) + j(Q) - 1$. The two possibilities are a downset contained in P , or all of P together with a nonempty downset of Q . These identities make (1.1) explicit for many recursively assembled alphabets.

9.3 A table of examples

In Table 1, $\rho(P) = n + j(P) - 2$ is the finite exponent parameter in $\omega^{\omega^{\rho(P)}}$.

Alphabet P	n	$j(P)$	$\rho(P)$	Maximal order type
Two-element chain	2	3	3	ω^{ω^3}
Two-element antichain	2	4	4	ω^{ω^4}
Three-element chain	3	4	5	ω^{ω^5}
Three-element V or its dual	3	5	6	ω^{ω^6}
Two-element chain plus an isolated point	3	6	7	ω^{ω^7}
Three-element antichain	3	8	9	ω^{ω^9}
Four-element chain	4	5	7	ω^{ω^7}
Four-element diamond	4	6	8	ω^{ω^8}
Four-element antichain	4	16	18	$\omega^{\omega^{18}}$
Five-element antichain	5	32	35	$\omega^{\omega^{35}}$

Table 1: Exact consequences of Theorem 1.1. The diamond has one bottom element, two incomparable middle elements, and one top element.

The alphabet cardinality by itself is not enough: the two binary values already differ. Conversely, nonisomorphic alphabets can have the same maximal order type. For instance the three-element V and its dual have the same n and j , while a four-element chain and a three-element chain plus an isolated point have the same value of $n + j - 2$ despite different cardinalities.

9.4 Finite quasi-orders

For a finite quasi-order rather than a poset, first pass to its quotient by mutual comparison. Projection of labels preserves and reflects word embedding: a comparison of quotient labels is exactly the comparison of any representatives. Therefore the theorem applies with n equal to the number of equivalence classes and j equal to the number of downsets of the quotient poset. Counting raw labels instead of classes can give an incorrect answer.

10 An exact symbolic embedding algorithm

10.1 Representation

The supplied module `code/omega2.py` represents a word by a finite sequence of tokens. A token (f', x) is a finite letter, and (w', D) is B_D , with D stored as a nonempty bitmask downset. The poset relation is supplied by predecessor masks. The code validates reflexivity, antisymmetry, and transitivity.

This representation does not claim to compute the recurrent set of an arbitrary black-box infinite input. Rather, it decides embeddings between already supplied finite periodic-block descriptions. The mathematical normal-form theorem says such descriptions represent every equivalence class in the target order.

10.2 Greedy decision procedure

Maintain a pointer to a target token, initially the first. Read source tokens from left to right.

Source token is a finite letter a .

Scan target tokens until finding either a finite letter b with $a \leq_P b$, or an infinite token B_E with $a \in E$. In the finite case consume the letter and advance the pointer. In the infinite case leave the pointer at B_E : after finitely many matches, an equivalent periodic tail remains.

Source token is an infinite block B_D .

Scan target tokens until finding an infinite token B_E with $D \subseteq E$. Consume its tail cofinally and advance the pointer to the following token.

If the required token does not exist, reject. Accept after every source token has been processed.

Theorem 10.1 (Exactness of the algorithm). *The greedy procedure accepts exactly when the represented source ordinal word embeds into the represented target ordinal word.*

Proof. Within a periodic target block, removing any finite initial segment leaves an equivalent tail. Thus the symbolic state need not record a phase or a finite offset in that block.

For a finite source letter, the first eligible target token provides an early possible match. Choosing a later token cannot leave a more useful suffix. Within an infinite token, choosing an actual sufficiently early eligible occurrence leaves an equivalent periodic tail. Therefore a successful continuation after any possible match can also be made after the greedy match.

For an infinite source block B_D , Lemma 4.1 shows that every embedding must use some target block B_E cofinally with $D \subseteq E$. All later source positions must then lie after that target block. The greedy choice takes the first eligible such block. The entire source B_D embeds in its available tail by Lemma 3.1; allowing part of the source block to occur earlier cannot improve the endpoint beyond this first eligible block. The remaining target suffix after the greedy choice contains the suffix left by any later choice.

Inducting on the number of remaining source tokens proves completeness: whenever a completion exists, the greedy step preserves the existence of a completion. Conversely, each accepted finite step can be realized by an actual letter match in a finite token or a periodic tail, and each accepted infinite step by a cofinal embedding into the selected target block. Concatenating these realizations gives an ordinal embedding, so acceptance is sound. $\square$

The target pointer never retreats. With bit operations and relation lookup counted as unit cost, the decision procedure uses $O(|u|_{\text{tok}} + |v|_{\text{tok}})$ time and constant auxiliary space. For unbounded alphabet size, bitmask operations have their ordinary bit costs; validating masks and the poset incurs additional cost. The reference checker uses memoized exploration of every eligible target endpoint rather than the first endpoint.

10.3 Why finite expansion is not an adequate substitute

Replacing each B_D by M repetitions for a large integer M changes the problem. The absorption equivalence $aB_{\downarrow a} \simeq B_{\downarrow a}$ already fails for the direction from length $M + 1$ to length M after a uniform finite expansion. Conversely, finite approximations do not directly certify the cofinal-boundary restrictions needed for reflection of product maps. The supplied code therefore treats ω -blocks symbolically and uses the proven cofinal semantics exactly.

11 Computational checks and reproducibility

The included verification run passed 1,710,857 assertions. Its random seed is 20260919; all random checks supplement deterministic checks. The report is stored in `data/verification.json`. The recorded run used Python 3.13.5, and the code targets Python 3.10 or newer without third-party packages.

The exhaustive comparison of the two deciders uses all naturally labelled posets of sizes zero through three and every source/target token word of length at most two. The longer-word comparisons cover all naturally labelled posets of sizes two through four and use token words of length at most twelve.

For the product embeddings, every naturally labelled poset of sizes two through four is processed, introducing all nonempty downsets in cardinality order. There are 355 marker stages. At each stage the exhaustive test uses a fixed four-word component pool and all pairs of tuples of arities two and three. The seeded tests add tuples of arities two through five with component token lengths at most eight. In total these give 1,580,460 product-map comparisons. Separate padding tests allow all one-token words, including already universal words, so padding reflection is checked beyond the narrower family needed in the main induction.

Check	Number of assertions
Exhaustive greedy/reference agreement	38,107
Seeded longer-word greedy/reference agreement	9,800
Exhaustive proper/universal product maps	1,544,960
Seeded longer-word product maps	35,500
Padding reflection	55,150
Padding has nonzero limit length	4,024
Universal-only comparison formula	20,640
Universal-only normalization	2,660
Boundary, absorption, and regression cases	16
Total	1,710,857

Table 2: Executed symbolic checks. A “word length” in the test generator means token length, not ordinal length.

The two embedding implementations share the mathematical periodic-tail semantics. Agreement between them is a check of the greedy optimization, not an independent derivation of that semantics. Likewise, no collection of the finite assertions in Table 2 proves ω^{ω^4} to be the maximal order type. That conclusion rests on the unbounded family of product embeddings and the ordinal calculation.

11.1 Enumeration conventions

The auxiliary enumerator considers relations on $\{0, \dots, n-1\}$ whose strict comparisons respect the usual numerical order. Every finite-poset isomorphism class has such a labelling, but the enumeration is neither an enumeration of all labellings nor an isomorphism-free enumeration. The numbers of generated relations for $n = 0, 1, 2, 3, 4, 5$ are respectively

$$1, 1, 2, 7, 40, 357.$$

These counts were computed by the included program. Histograms by the number of downsets appear in `data/ideal_counts.csv`. Table examples are also exported to `data/examples.csv`. No claim about uniform sampling of isomorphism classes is intended.

To compute $j(P)$ for one small input poset, the code simply checks all 2^n subsets for downward closure. This exponential enumeration is sufficient for the examples and the proof tests. The article does not claim a polynomial-time algorithm for downset counting.

11.2 Reproduce the package

From the root of the unpacked directory:

```
python3 code/verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Python command regenerates the verification reports and CSV tables. The PDF can alternatively be built by running `pdflatex article.tex` repeatedly until cross-references stabilize. No external bibliography processor is required; a separate BibTeX file is supplied for reuse.

12 Proof audit, scope, and further directions

12.1 Where a hidden gap would have to occur

The proof is organized so its nonclassical burden is concentrated in a few claims. The atom-count upper bound is a monotone-surjection argument. Cofinal localization uses finiteness of the target token

decomposition. Synchronization uses the newness condition $D \not\subseteq E$ for all old blocks, not merely distinctness of block labels. The proper-marker reflection uses downward closure to exclude a guard from the next marker. The full-marker reflection uses *both* cancellation and the limit length of the padded component.

These dependencies explain several tempting but incorrect shortcuts. A source infinite block need not land wholly inside one target token. The ordinary evaluation map from atom strings need not reflect order. The universal marker admits no excluded finite letter. A finite test cannot certify an infinite supremum. None of these shortcuts is used.

An independent referee can focus first on Theorems 5.2 and 6.3. Once their product-reflection assertions are accepted, the classification follows by a short finite induction and the classical finite-product formula. The binary specialization in Section 8 provides a second presentation of the same core argument, not an independent proof with different foundations.

12.2 Established within the argument versus not claimed

Within this manuscript, the main theorem has a complete proof relative to the classical results stated in Theorem 2.2. There is no remaining conjectural lemma inside that derivation. Nevertheless the argument has not undergone independent expert review or proof-assistant verification. The appropriate external status is therefore a *proposed solution*, with explicit hypotheses and reproducible supporting checks, rather than a certified resolution.

The article does not compute all invariants of these quasi-orders: their height, width, and exact isomorphism classification are different tasks. It does not show that two alphabets with equal $n + j(P)$ produce isomorphic word orders. It does not give a canonical representation for arbitrary infinite input streams. It also does not assert that selected block families at higher ordinal cutoffs have the same counting law.

12.3 Why the proof stops at this cutoff

The decisive compactness feature is that a target word below ω^2 has only finitely many ω -blocks. The image of a source ω -block must eventually stabilize in one of them. At length ω^2 , a source ω -block may instead visit infinitely many successive target ω -blocks, taking only finitely many points from each. Thus the cofinal localization lemma in its present form is no longer valid when that target length is allowed.

A higher-cutoff extension would need a hierarchy of separator ranks and a new localization theorem that accounts for escape through infinitely many lower-rank blocks. Counting indecomposable blocks without that theorem would reproduce the same logical mistake that naive atom counting makes at the universal-only stage.

For infinite alphabets, even when every individual word has finite image, the set of possible recurrent downsets need not be finite, and a single finite inclusion ordering does not exhaust the construction. The present finite induction therefore does not directly transfer. These observations identify precise missing mechanisms for extensions; they are not reasons to reject further investigation.

12.4 A concrete formal-verification target

A useful formalization would first define ordinal concatenation and its embedding relation for finite periodic-token codes, then prove the exact symbolic decision theorem. The next targets are the cofinal boundary lemma, proper-marker reflection, and universal-marker reflection. With the classical maximal-order-type product and finite-word theorems available, the final induction has very little additional complexity. The package contains executable tests but no Lean development, and it makes no claim of machine-checked proof.

A A compact proof-dependency map

Statement	Inputs	Role
Finite representation, Proposition 3.3	Finiteness of P ; recurrent labels; monotone concatenation	Identifies the full word quotient with a finite-token quotient.
Atom-count upper bound, Proposition 3.5	Finite-word formula; monotone surjection	Gives $o(\mathcal{W}(P, \mathcal{F})) \leq H_{n+ \mathcal{F} }$.
Cofinal localization, Lemma 4.1	Finitely many target tokens; strict increase; periodic recurrence	Finds the target block used cofinally by each source marker.
Synchronization, Lemma 4.3	Cofinal localization; no old block contains the new downset	Forces the i th new marker to correspond to the i th new marker.
Proper product embedding, Theorem 5.2	Synchronization; a letter outside the proper downset	Embeds every finite Cartesian power of the old stage.
Limit padding, Lemma 6.1	A proper old block; an excluded letter; final-letter cancellation	Produces an order-reflecting map into nonzero limit lengths.
Universal product embedding, Theorem 6.3	Synchronization; limit padding; the finite-bound contradiction	Adds the universal block without losing the product lower bound.
Family classification, Theorem 7.1	The two product maps; classical natural product; atom upper bound	Computes every selected family, including its sole exception.
Main theorem, Theorem 1.1	Full family has a proper downset when $n \geq 2$; finite representation	Gives $\omega^{\omega^{n+j(P)-2}}$ and the two boundary cases.

B Minimal worked use of the code

The following example can be run with `code/` on the Python import path. Downset masks 1, 2, and 3 represent $\{a\}$, $\{b\}$, and $\{a, b\}$.

```

from omega2 import FinitePoset, letter, omega, embeds, theorem_value

P = FinitePoset.antichain(2)
a, b = letter(0), letter(1)
A, B, U = omega(1), omega(2), omega(3)

assert embeds(P, (a, U), (U,)) # finite-prefix absorption
assert embeds(P, (A, ), (U,)) # pure tail into universal tail
assert not embeds(P, (A, B), (U,)) # two cofinal blocks cannot share one
assert not embeds(P, (U, ), (A, B)) # no eligible cofinal destination

print(theorem_value(P))
# omega^(omega^4)

C = FinitePoset.chain(2)
print(theorem_value(C))
# omega^(omega^3)

print(theorem_value(P, family=(P.full,)))
# omega^(omega^1 + 1) = omega^(omega + 1)

```

The last output illustrates the exceptional universal-only family. Its maximal order type is much smaller than $H_3 = \omega^{\omega^2}$, even though the two finite letters and the universal block form three distinct formal tokens.

C Literature and reproducibility record

The chosen source was checked both in arXiv HTML and in the PDF rendering; Example 3.15 is on printed page 10, which is zero-based PDF page 9. Its version identifier is 2409.03199v2. The source of the open question, the two standard maximal-order-type references, and the distinction between proven classical inputs and the present proposed argument are recorded in `notes/literature_audit.md`.

The archive contains this source and its compiled PDF, a reusable `references.bib`, the symbolic module, the reproducible verification script, machine-readable test results, example and downset-count tables, a proof-audit note, and a README. It contains no third-party article PDFs, font files, checksum files, or LaTeX build intermediates.

References

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